(a) Mapping (cf. paper Fig. 8)
Seq.: MLP-centric
Seq.: Locality-centric
Stride: MLP-centric
Stride: Locality-centric
paper: ~0.30
1.00
0.17
1.00
0.50
0.0
0.2
0.4
0.6
0.8
1.0
DRAM BW (norm. to MLP-centric)
(b) HetMap memcpy (cf. paper Fig. 14)
4C-16R PIM-MMU
4C-16R Baseline
4C-8R PIM-MMU
4C-8R Baseline
2C-4R PIM-MMU
2C-4R Baseline
4.3x (31 GB/s)
1.0x (7 GB/s)
4.3x (31 GB/s)
1.0x (7 GB/s)
2.3x (16 GB/s)
1.0x (7 GB/s)
0
2
4
6
DRAM memcpy throughput (norm.)